PictoBlox

CODEAVOUR 7.0
INTERNATIONAL

# DR. TITAN
## AI MENTAL HEALTH & FITNESS CHECK ROBO

AI POWERED MENTAL WELLNESS & FITNESS ANALYSIS ROBOT









# INTRODUCTION

In the modern world, technology is advancing at an extraordinary pace, transforming how we live, work, and communicate. Despite these advancements, one of the most important aspects of human life — **mental and emotional well-being** — is often overlooked. While people actively monitor physical health and productivity, emotional health remains a silent concern affecting individuals across all age groups.

Today, students face academic pressure, professionals deal with stress, and elderly individuals often experience loneliness. These challenges are not always visible, yet they deeply impact overall well-being. In many cases, people hesitate to express their emotions or do not have access to continuous support systems.

To address this growing concern, **DR. TITAN (Therapeutic Intelligence and Training Assistance Network)** has been developed as an innovative solution that combines **Artificial Intelligence (AI), robotics, and smart health monitoring technologies** into one integrated system.

DR. TITAN is designed to function as an **intelligent companion** capable of:

- Monitoring physical health parameters
- Understanding human emotions using AI
- Interacting naturally with users
- Providing real-time support and guidance
- Ensuring personal safety through advanced systems

The core philosophy of DR. TITAN is simple yet powerful:
*Technology should not just perform tasks — it should understand, support, and care for human life.*


AI MENTAL HEALTH
AND FITNESS CHECK ROBO
Smart Intelligence For Mind & Body Well-being
ABOUT THE PROJECT
An advanced AI-powered healthcare system designed to monitor, analyze, and support mental health and physical fitness.
This intelligent solution combines emotional well-being analysis with fitness and vital parameter monitoring to provide real-time insights and personalized wellness guidance, ensuring a healthier and more balanced lifestyle.
SYSTEM OVERVIEW
The AI Mental Health and Fitness Check Robo continuously evaluates mental and physical health indicators using intelligent data interpretation. It assists individuals by identifying stress patterns, emotional states and fitness levels while offering smart recommendations for improvement.
KEY CAPABILITIES
Mental Health Pattern Analysis
Identifies emotional trends and mental well-being indicators through AI-driven
Stress & Emotional State Detection
Monitors stress levels and emotional fluctuations using behavioral and biomettic data.
Fitness & Vital Parameter Monitoring
Tracks heart rate, activity levels, and essential fitness metrics in real time.
Personalized Wellness Insights
Provides customized health suggestions based on individual data patterns.
Continuous AI-Driven Support
Ensures 24/7 intelligent assistance for -.*... both mental and physical health monitoring.
Fitness & Vital Parameter Monitoring
Tracks heart rate, activity levels, and essential fitness metrics in real time.
PLICATION AREAS
& Wellness Centers
Educational Institutions
Personal Health Management
Team Details
Team Rockerz
Presented By:
Mohammed Hasen Khan
" Whe Intelligence Meets Human Well-Being "
2025-26
CODEAVOUR 7.0
INTERNATIONAL
Track 1: Innovation and Entrepreneurship
Rockerz
Dr TITAN - AI Mental Health
Fitness Check Robo
ect: An AI-powered health
physical and mental support.
ne monitoring and intelligent
interaction.
DR.TITAN
AI MENTAL HEALTH &
FITNESS CHECK ROBO

# CREATOR DETAILS

The DR. TITAN project has been conceptualized and developed by **Mohammed Hasen Khan**, a passionate young innovator with a strong vision to use technology for societal betterment.

As a student of Grade 7, he has demonstrated exceptional talent in multiple domains, including **public speaking, innovation, leadership, and creative problem-solving**. His ability to think beyond conventional boundaries and address real-world challenges has led to the creation of this impactful project.

🔷 **Role and Contributions:**

- Designed the complete concept of DR. TITAN
- Integrated hardware components and system architecture
- Developed AI-based functionalities using PictoBlox
- Implemented sensor-based monitoring systems
- Worked on emotional detection and interaction logic
- Prepared documentation and presentation

◇ **Achievements and Recognition:**

- Awarded by India Book of Records for multilingual speech delivery
- Recognized as a “Master Orator” for excellence in public speaking
- National-level gold medalist in speech competitions
- Selected to represent at international-level competition (Codeavour 7.0)
- Multi-talented achiever in sports, martial arts, and creative platforms

His vision is to create solutions that not only demonstrate innovation but also bring **real change in people’s lives**. DR. TITAN stands as a reflection of this vision.

## THEME OF THE PROJECT

The theme of this project is:

**“Technology for Mental Health, Physical Well-being, and Human Safety”**

This theme highlights the importance of using modern technologies such as:

- Artificial Intelligence (AI)
- Internet of Things (IoT)
- Robotics and automation

to solve real-life problems related to:

- Emotional health
- Physical fitness
- Personal safety

The project emphasizes the concept of **preventive care**, where issues are identified and addressed before they become serious. It also promotes the idea that technology should be **accessible, supportive, and human-centered**.

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## REASON FOR SELECTING THE PROJECT

The selection of this project is based on the growing need to address critical challenges in society related to mental health, health monitoring, and personal safety.

In today’s world:

- Stress levels among students are increasing due to academic pressure
- Many individuals struggle to express their emotions
- Elderly people often face loneliness and lack companionship
- Health issues are detected late due to lack of continuous monitoring

- Existing safety systems depend on manual action, which may not always be possible

These challenges highlight the need for a **smart, automated, and intelligent system** that can continuously monitor, understand, and respond to human needs.

DR. TITAN was chosen as a project to:

- Develop a **human-centered AI solution**
- Learn and implement real-world technologies
- Create a system that is **practical, scalable, and impactful**
- Contribute to improving the quality of human life

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## OVERVIEW OF THE SYSTEM

DR. TITAN is a **multi-functional AI-powered humanoid system** designed to integrate health monitoring, emotional intelligence, and safety features into one platform.

The system works by continuously collecting data from sensors, analyzing it using AI, and responding intelligently based on the situation.

It acts as:

- A **health assistant**
- An **emotional companion**
- A **safety monitoring system**

This integration makes DR. TITAN a **comprehensive solution** rather than a single-purpose device.

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## WORKING PRINCIPLE

The working of DR. TITAN is based on three main stages:

### 🔷 1. Data Collection

The system collects real-time data using various sensors and input devices. These include:

- Heart rate using pulse sensors
- Facial expressions using a camera
- Environmental and activity data through sensors

This continuous data collection ensures that the system is always aware of the user's condition.

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### 🔷 2. AI-Based Analysis

The collected data is processed using Artificial Intelligence. The system analyzes:

- Emotional expressions
- Stress patterns
- Health indicators

AI helps the system understand not just physical data but also **emotional and behavioral patterns**, making it more intelligent and responsive.

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### 🔷 3. Intelligent Response

Based on the analysis, DR. TITAN provides appropriate responses such as:

- Talking to the user
- Giving health suggestions
- Playing calming music
- Providing motivation
- Triggering alerts if necessary

This creates a **complete interaction loop**, making the system dynamic and user-friendly.

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# KEY FEATURES OF DR. TITAN

DR. TITAN is designed as a multi-functional system that integrates several advanced features to provide complete support in terms of **health, emotional well-being, and safety**.

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### Mental Health Detection System

One of the most powerful features of DR. TITAN is its ability to understand human emotions using Artificial Intelligence.

The system uses a camera along with AI models to:

- Detect facial expressions
- Identify emotional states such as happiness, sadness, stress, and anger
- Analyze behavioral patterns over time

This allows the system to recognize early signs of mental stress or emotional discomfort. Unlike humans, who may hesitate to ask questions, DR. TITAN continuously observes and responds without judgment.

This feature helps:

- Reduce emotional isolation
- Encourage expression
- Provide early support for mental health issues

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### Physical Health Monitoring System

DR. TITAN continuously monitors important health parameters using sensors. It collects data such as:

- Heart rate using a pulse sensor
- Body metrics like height and weight for BMI calculation
- Daily activity levels

The system analyzes this data and provides feedback regarding the user's health condition. It helps users:

- Understand their physical health
- Maintain a balanced lifestyle
- Detect abnormalities early

This feature promotes **preventive healthcare**, reducing the chances of serious health issues.

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### AI-Based Conversation System

DR. TITAN is not limited to mechanical responses. It uses AI-based conversational systems to interact naturally with users.

The robot can:

- Talk and listen to users
- Respond intelligently to queries
- Motivate and support users
- Engage in meaningful conversations

This creates a sense of companionship, especially for individuals who feel lonely or stressed. The interaction is dynamic and adaptive, making it more engaging and realistic.

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### Mental Wellness Support System

When the system detects stress or emotional imbalance, it activates its wellness support features.

These include:

- Playing calming or meditation music
- Speaking in a soothing tone
- Suggesting relaxation techniques

This helps users:

- Calm down during stressful situations
- Improve emotional stability
- Maintain mental peace

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**Elderly Care Support System**

DR. TITAN plays an important role in supporting elderly individuals who often face loneliness and health-related challenges.

The system provides:

- Medicine reminders
- Friendly interaction and conversation
- Emergency assistance features

This helps elderly users feel:

- Connected
- Safe
- Supported

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**Student Support System**

Students often face high levels of stress due to academic pressure. DR. TITAN helps by:

- Detecting stress levels
- Providing motivation
- Encouraging positive thinking

It acts as a **non-judgmental companion**, helping students maintain focus and emotional balance.

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**Posture Detection System**

Incorrect posture is a common issue, especially among students and professionals.

DR. TITAN uses AI-based body detection to:

- Monitor posture
- Identify incorrect sitting or standing positions

- Provide corrective suggestions

This helps prevent long-term health problems such as back pain and spinal issues.

## ADVANCED SAFETY SYSTEM – SURAKSHANET

SurakshaNet is an advanced feature integrated into DR. TITAN that focuses on **personal safety and continuous monitoring**.

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**Introduction**

Traditional safety systems depend on manual activation (SOS buttons), which may not be effective in real-life emergencies. SurakshaNet overcomes this limitation by providing **automatic and continuous safety tracking**.

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**Problem Addressed**

In many emergency situations:

- The victim may be unconscious
- Unable to use a phone
- Under threat or fear

In such cases, manual systems fail, leading to delayed response and increased risk.

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**Solution**

SurakshaNet provides:

- Continuous location tracking
- Automatic alert generation
- Real-time communication with trusted contacts

This ensures that help can be provided even when the user cannot act.

## Working Principle

The system operates through:

- GPS module → tracks real-time location
- ESP32 → processes data
- GSM/IoT → sends alerts and updates

Location data is continuously shared with:

- Parents
- Guardians
- Emergency contacts

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## Key Innovations

- Fully automatic system
- No dependency on SOS button
- Real-time tracking
- Works in all environments

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## Applications

- Women safety
- Student tracking
- Child protection
- Elderly monitoring

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## Impact

SurakshaNet significantly improves safety by:

- Reducing response time
- Increasing reliability

- Ensuring continuous protection

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## HARDWARE COMPONENTS

DR. TITAN is built using a combination of electronic components that work together to perform different functions.

Key components include:

- **Arduino Uno** – Main controller for processing inputs and outputs
- **ESP32** – Enables Wi-Fi connectivity and IoT communication
- **Pulse Sensor** – Measures heart rate
- **Ultrasonic Sensor** – Detects nearby objects
- **Webcam** – Captures facial data for AI processing
- **Sound Sensor** – Detects sound and enables interaction
- **LDR Sensor** – Detects light intensity
- **Servo Motors** – Controls movement of robot parts
- **Gear Motors** – Used for mobility
- **Motor Driver** – Controls motor operations
- **LCD Display** – Displays information
- **GPS Module** – Tracks location (SurakshaNet)
- **GSM Module** – Sends alerts and messages
- **Battery (Li-Po)** – Powers the system

Each component plays a crucial role in ensuring smooth and efficient functioning.

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## SOFTWARE USED

The system is developed using:

🔷 **PictoBlox**

- Used for AI integration
- Supports face detection and emotion recognition
- Allows block-based and Python coding

🔷 **Arduino IDE**

- Used for programming hardware components
- Controls sensors and actuators

🔷 **AI Conversational System**

- Enables interaction between user and robot
- Provides intelligent responses

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## APPLICATIONS

DR. TITAN can be applied in multiple real-world scenarios:

- **Healthcare** – Monitoring patients and providing support
- **Education** – Helping students manage stress
- **Elderly Care** – Providing companionship and safety
- **Homes** – Acting as a personal assistant
- **Smart Cities** – Integrating safety and monitoring systems

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## SDG ALIGNMENT

DR. TITAN supports global development goals:

- **Goal 3: Good Health & Well-being**
  Promotes physical and mental health
- **Goal 4: Quality Education**
  Supports students emotionally
- **Goal 5: Gender Equality**
  Enhances safety for women

- **Goal 9: Innovation & Infrastructure**
  Encourages technological advancement
- **Goal 16: Peace & Safety**
  Improves personal security

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## ADVANTAGES

DR. TITAN offers several advantages:

- Early detection of health and emotional issues
- Continuous monitoring and support
- Reduces loneliness
- Enhances personal safety
- Encourages healthy lifestyle
- Provides real-time assistance

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## FUTURE SCOPE

The future scope of DR. TITAN is vast and promising.

In future, the system can be enhanced by:

- Developing a mobile application
- Adding voice-based AI assistants
- Integrating with smart home systems
- Improving AI accuracy for emotion detection
- Adding wearable device support
- Cloud-based health reports

These improvements can make DR. TITAN more powerful and widely usable.

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## CONCLUSION

DR. TITAN represents a significant step forward in the field of **human-centered technology**.

It is not just a robot or a project — it is a **complete system designed to understand, support, and protect human life**.

By combining:

- Health monitoring
- Emotional intelligence
- Safety systems

DR. TITAN creates a solution that is both **innovative and meaningful**.

It addresses real-world problems that are often ignored and provides a practical approach to improving quality of life.

In a world where people often feel unheard and unsupported, DR. TITAN ensures that:

- No emotion goes unnoticed
- No health issue is ignored
- No person feels alone

*DR. TITAN is not just an innovation -*

*It is a necessity for a world that needs smarter care, deeper understanding, and continuous protection.*